

Manish Rao

+1 (512) 829 -1911 | manishrao.dev9@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | Austin, TX | Open to Relocate

PROFESSIONAL SUMMARY

Senior Software Engineer with 5+ years experience building distributed backend systems and financial platforms using Java, Spring Boot, Node.js, and Python. Skilled in microservices, event-driven systems, REST/GraphQL APIs, frontend frameworks React, Angular. Experienced in AWS, Docker, Kubernetes, Terraform, CI/CD, observability Prometheus, Grafana, ELK, AI/ML integration with Hugging Face and RAG workflows.

PROFESSIONAL EXPERIENCE

JP Morgan Chase & Co. | *Software Engineer*

Dallas, Texas | Jan 2025 – Present

- Designed and implemented secure, high-throughput payment APIs using **Java, Spring Boot, and Spring Security** supporting ACH transfers, internal account settlements, and authorization workflows handling high-volume concurrent requests.
- Implemented **OAuth2 and JWT**-based authentication integrated with internal identity services and API gateway layers enabling secure token-based authorization across multiple financial microservices.
- Optimized **PostgreSQL schemas, indexing strategies**, and transactional queries for payment records and settlement logs improving reconciliation and transaction lookup performance by **35%** under high concurrency.
- Developed payment validation and fraud detection microservices integrating internal compliance systems and asynchronous Kafka event streams to analyze transaction patterns before payment authorization.
- Reduced payment incident triage **from 3+ hours to under 30 minutes** by building a **LangGraph** multi-agent investigation system with **LangChain RAG, surfacing runbooks, settlement logs**, and compliance policies in real time across **ACH infrastructure**, reducing on-call escalations by 25%.
- Applied **Redis-based caching** and idempotency strategies for payment session and account validation data reducing transaction processing latency from **350ms to under 120ms** during peak banking traffic.
- Containerized payment services using Docker and deployed on **AWS EKS** with Kubernetes autoscaling and Helm-based deployments integrated into **Jenkins CI/CD** pipelines enabling reliable releases and scalable infrastructure.
- Partnered with the data science team to build an internal AI-powered engineering assistant using Generative AI APIs, prompt engineering, and **Hugging Face** transformer models to securely query internal payment documentation reducing troubleshooting time by **25%**.

Accenture | *Software Engineer*

India | Jan 2021 – July 2023

- Constructed backend services for digital banking transactions, account management, and payment processing using Java, Spring Boot, Hibernate, and REST APIs, reducing API latency from **5s to under 2s** and improving reliability for high-volume banking operations.
- Orchestrated event-driven microservices using **Apache Kafka** enabling asynchronous processing of payment workflows, account updates, and notification pipelines improving throughput and service decoupling.
- Formed React.js dashboards using **TypeScript** integrated with Spring Boot APIs enabling real-time monitoring of payment processing, account activity, and fraud alerts for banking operations teams.
- Managed AWS deployments using **EC2, S3, and RDS** with Docker containers and **Terraform** infrastructure provisioning enabling automated environment setup and scalable deployments.
- Formulated and optimized **PostgreSQL** database schemas and indexing strategies for transaction history and reconciliation modules improving batch processing and reporting performance.
- Established JWT authentication and role-based access control for secure banking APIs and standardized API documentation using OpenAPI/Swagger improving service discoverability across teams.
- Strengthened service reliability by implementing automated tests using **JUnit** and **Mockito** and introducing application monitoring using **Prometheus** and Grafana dashboards.
- Provided mentorship and technical guidance to junior engineers, conducting code reviews, knowledge-sharing sessions, and pairing on complex microservices tasks to improve team productivity and code quality.

Tatvasoft | *Software Engineer*

India | Jul 2019 – Dec 2020

- Architected backend services using **Node.js** and Express.js for a supply chain platform implementing **REST APIs** for order processing, shipment tracking, and inventory updates supporting **5,000+** monthly transactions.
- Built responsive front-end components using **React.js**, JavaScript, and Bootstrap improving dashboard usability and reducing page load time by **20%** for logistics and warehouse operations teams.
- Engineered and optimized **MySQL** database schemas, queries, and stored procedures improving query performance by **25%** and enabling reliable synchronization across warehouse systems.
- Integrated third-party logistics APIs & implemented data validation workflows improving shipment tracking accuracy & reducing operational updates.
- Collaborated in Agile development cycles using **Jira** for sprint planning, backlog tracking, code reviews, and Git-based workflows contributing to bi-weekly releases and resolving production issues.

TECHNICAL SKILLS

Programming Languages: Java, Python, Golang, JavaScript, TypeScript, SQL

Backend & Frameworks: Spring Boot, Spring MVC, Hibernate, Node.js, Express.js, RESTful APIs, GraphQL, gRPC

Frontend & UI: React.js, Angular, HTML5, CSS3, Tailwind CSS, Bootstrap, Redux State Management

Distributed Systems: Microservices Architecture, Event-Driven Architecture, Apache Kafka, RabbitMQ, Apache Flink, Apache Spark

Databases & Storage: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch

Cloud Infrastructure & DevOps: AWS (EC2, S3, RDS, EKS), Docker, Kubernetes, Helm, Terraform, Jenkins, CI/CD Pipelines

Observability & Monitoring: Prometheus, Grafana, ELK Stack, Splunk, AWS CloudWatch, OpenTelemetry, Distributed Tracing

AI/ML Integration: Generative AI APIs, Hugging Face Transformers, Retrieval-Augmented Generation (RAG), Vector Search, LangChain, LangGraph

API & Security: OAuth2, JWT Authentication, RBAC, API Gateway, OpenAPI/Swagger, API Versioning

Engineering Practices: System Design, Data Structures & Algorithms, Distributed Systems Design, Agile/Scrum, Code Reviews

EDUCATION

Master of Science in Software Engineering, Northeastern University, Boston, MA

Sep 2023 – May 2025

Bachelor of Engineering in Computer Science, N.M.A.M. Institute of Technology, Nitte, Karnataka, India

Aug 2016 – Jul 2020

ACHIEVEMENTS & CERTIFICATIONS

Award: Accenture Celebrates Excellence (ACE) Award

Teaching: TA for CS7380 (Theory & Practical Applications of Generative AI)

Certifications: Neo4j Professional | Neo4j Graph Data Science | Intro to LangGraph | Agentic Knowledge Graph Construction

Pydantic for LLM Workflows | Building AI Voice Agents for Production | Evaluating AI Agents | Intro to Agent Observability & Evaluations